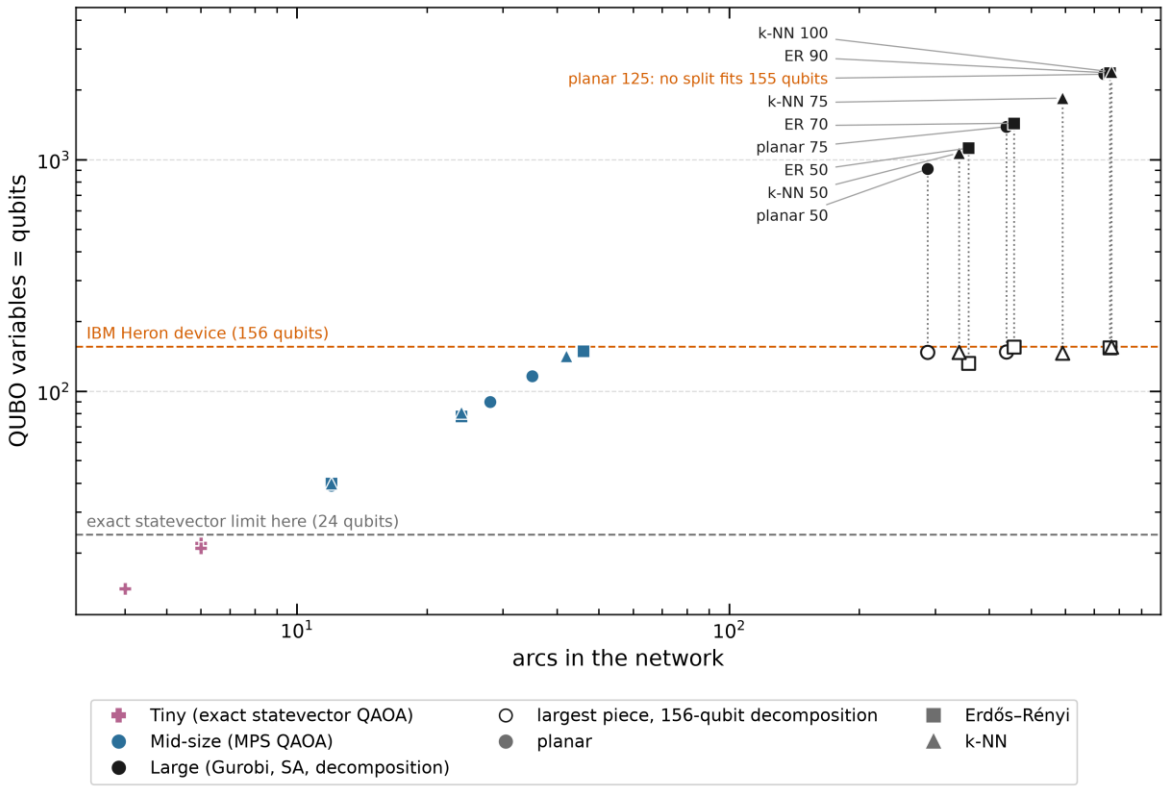
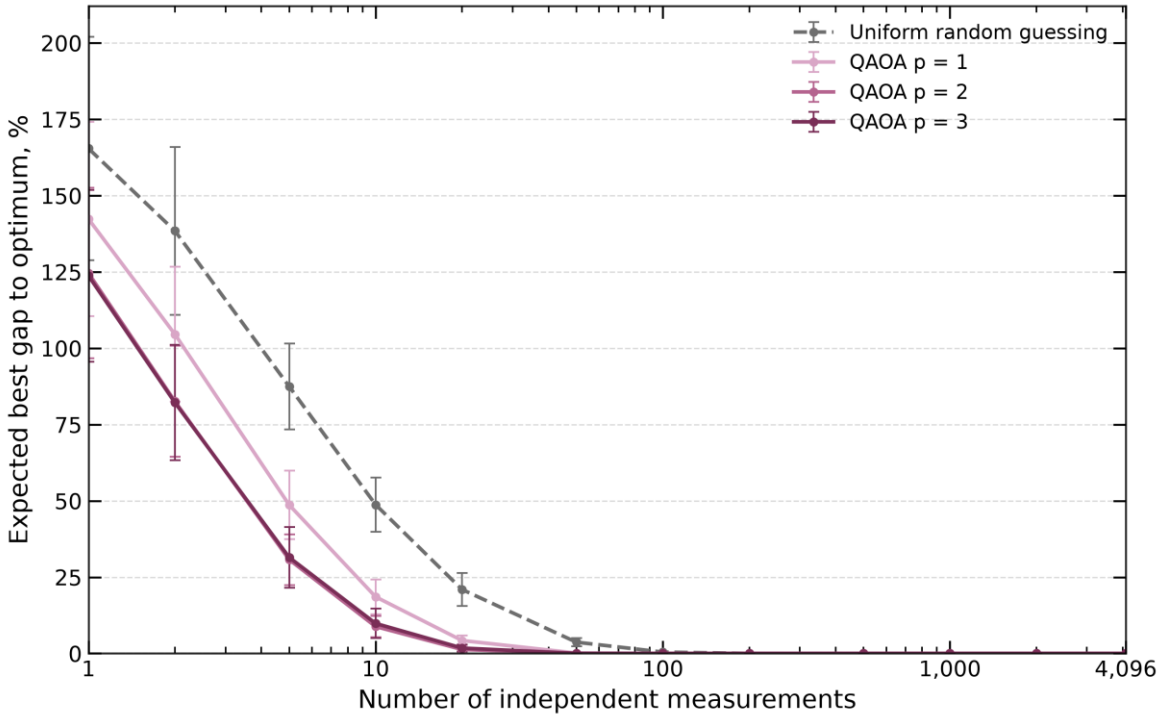
Implementing Quantum Approximate Optimization Algorithm

Remove links under a fixed budget to minimize the flow that remains.

Problem sizes and device limits



Expected best cut after N measurements

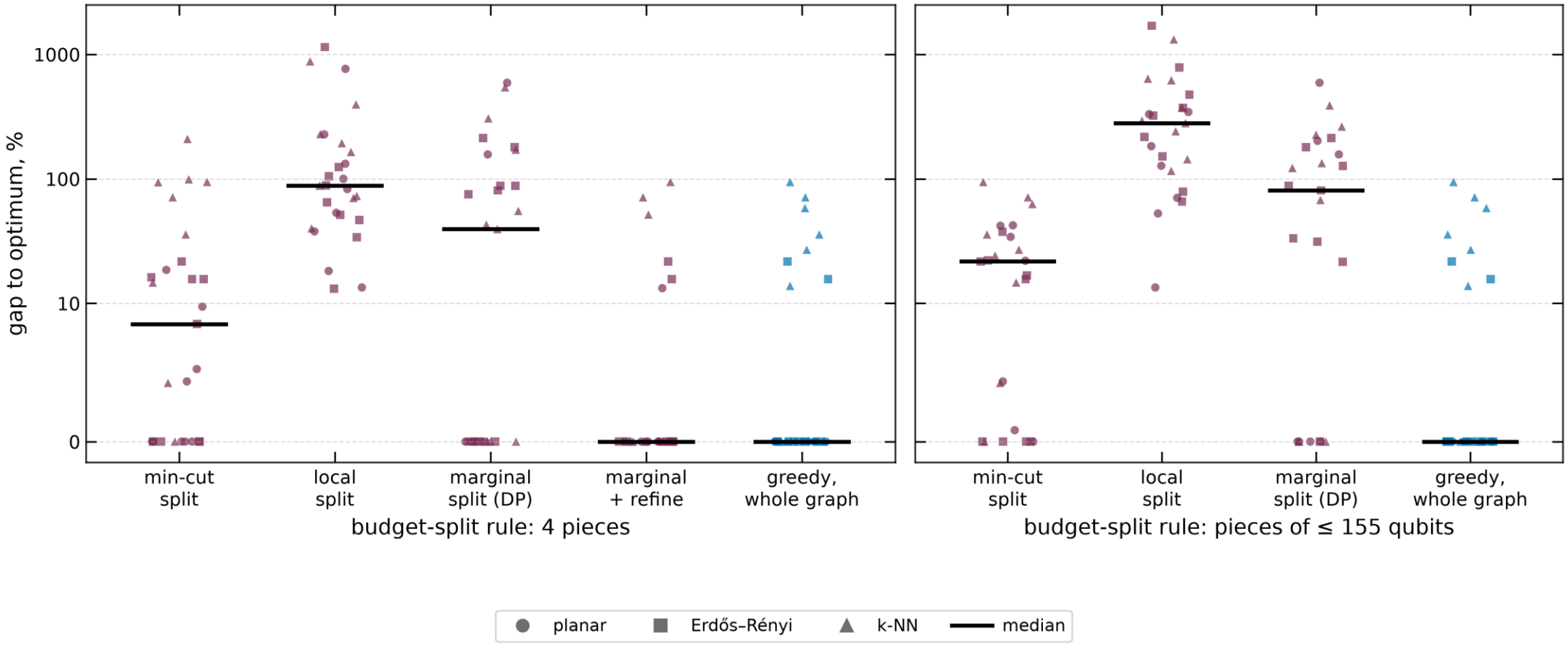


10 measurements: 9% gap (depth 2), 49% (random).
Statevector prediction; 6 tiny cases; mean ± SEM.

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Decomposition-Based Refinement

Validate penalties → partition → allocate budget → solve pieces → refine cuts

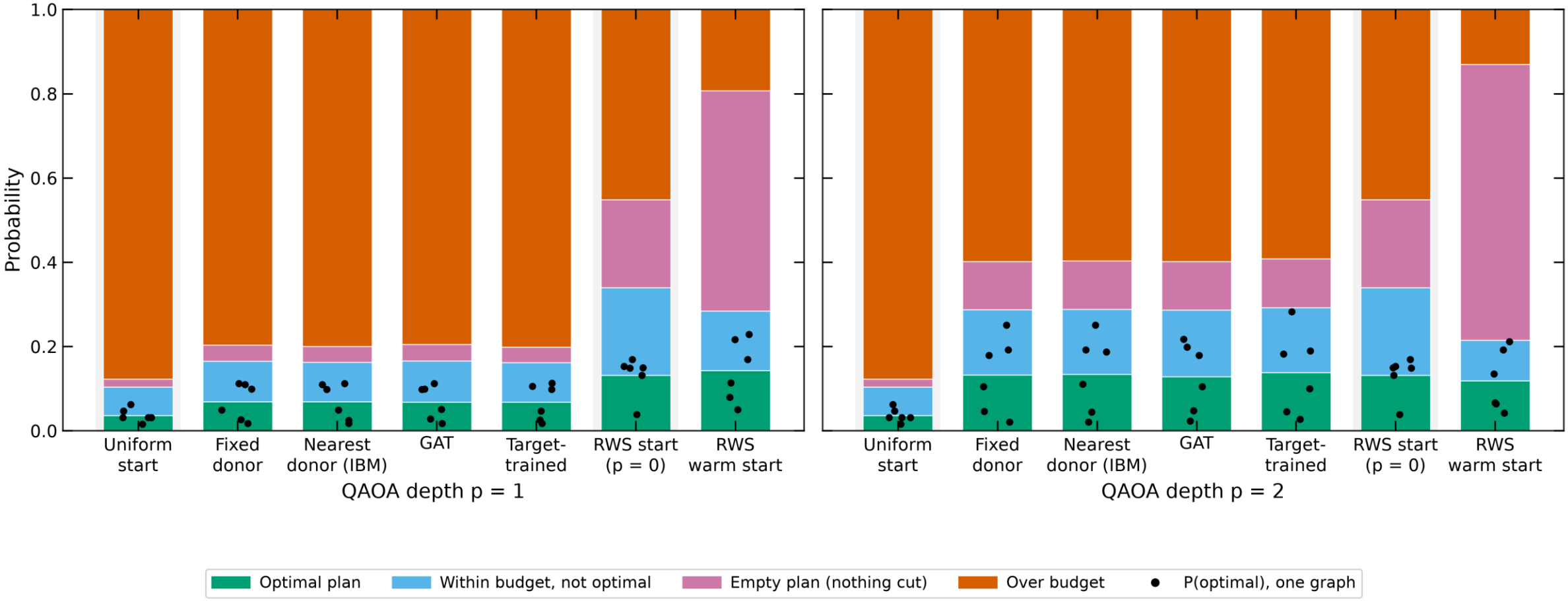


Four pieces + refinement: optimal in 21/27 cases. Device-sized pieces: 22% median gap with the best split rule.

27 cases, 50–125 nodes; exact piece solves. Gurobi solves all quickly; SA + repair never beats greedy.

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Parameter Setting Strategies



Fixed, nearest, GAT and trained angles give similar outcomes.
At p = 2, RWS assigns 66% probability to the empty plan.

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